HIllClimbing output:

```
Restart 5: Initial state = [1, 1, 0, 2]
Initial cost = 2

Current state: [1, 1, 0, 2]
Current cost: 2
Neighbor states and costs:
 Move column 0 to row 0: [0, 1, 0, 2], cost = 3
 Move column 0 to row 2: [2, 1, 0, 2], cost = 4
 Move column 0 to row 3: [3, 1, 0, 2], cost = 1
 Move column 1 to row 0: [1, 0, 0, 2], cost = 3
 Move column 1 to row 2: [1, 2, 0, 2], cost = 2
 Move column 1 to row 3: [1, 3, 0, 2], cost = 0
 Move column 2 to row 1: [1, 1, 1, 2], cost = 4
 Move column 2 to row 2: [1, 1, 2, 2], cost = 3
 Move column 2 to row 3: [1, 1, 3, 2], cost = 3
 Move column 3 to row 0: [1, 1, 0, 0], cost = 3
 Move column 3 to row 1: [1, 1, 0, 1], cost = 5
 Move column 3 to row 3: [1, 1, 0, 3], cost = 3

GOAL FOUND!
Final board: [1, 3, 0, 2]
Final cost: 0

Grid Representation:
. . Q .
Q . . .
. . . Q
. Q . .
```